Q1: Where is DDR register located (for each port)?

The Data Direction Register (DDR) is located physically inside the microcontroller chip and logically within its I/O memory space.

Logically (Memory Map Locations)

For each physical port (a grouping of up to 8 pins), there is a corresponding DDR. In the ATmega328P memory map, they are assigned the following **specific addresses:**

Port B (DDRB): * I/O Address: 0x04

SRAM Data Memory Address: 0x24

Controls pins typically associated with digital pins 8 through 13.

Port C (DDRC):

I/O Address: 0x07

SRAM Data Memory Address: 0x27

Controls pins typically associated with analog input pins A0 through A5 (which can also be used as digital I/O).

Port D (DDRD):

I/O Address: 0x0A

SRAM Data Memory Address: 0x2A

Q2: SRAM Vs Flash?

1.Flash Memory (The Program Memory)

Because it retains data without power, Flash is where your compiled code (your .hex or .bin file) lives. When you plug an Arduino or custom board into your computer and click "Upload," you are flashing the program into this memory.

Execution: In Harvard architecture microcontrollers (like AVR or ARM Cortex-M), the CPU fetches its instructions directly from Flash memory.

Constant Data: You can also use Flash to store unchanging data, like large lookup tables or text strings, so they don't consume precious SRAM.

2. SRAM (The Data Memory)

SRAM is built using flip-flop circuitry (typically 6 transistors per memory cell). "Static" means it doesn't need to be constantly refreshed to keep its data (unlike the DRAM in your laptop), making it incredibly fast and reliable for the CPU to access.

As soon as your microcontroller boots up, it uses SRAM to:

Store Global/Static Variables: Any variable whose value will change during runtime is placed here.

Manage the Stack: It keeps track of function calls, interrupts, and local variables.

Manage the Heap: If you use dynamic memory allocation (malloc() in C), it takes up SRAM.